

Android Application Development Lab

Beginner level Experiments

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Q1 -> Create an application that takes the name from a text box and shows hello message along with the name entered in text box, when the user clicks the OK button.

Ans1 ->

Xml -

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp"
    android:gravity="center">

    <EditText
        android:id="@+id/etName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter your name"
        android:inputType="textPersonName"/>

    <Button
        android:id="@+id/btnOK"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:text="OK"/>

    <TextView
        android:id="@+id/tvHello"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="24dp"
        android:textSize="20sp"
        android:textStyle="bold"
        android:text=" " />
```

</LinearLayout>

Java -

```
package com.example.a1_user_input;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

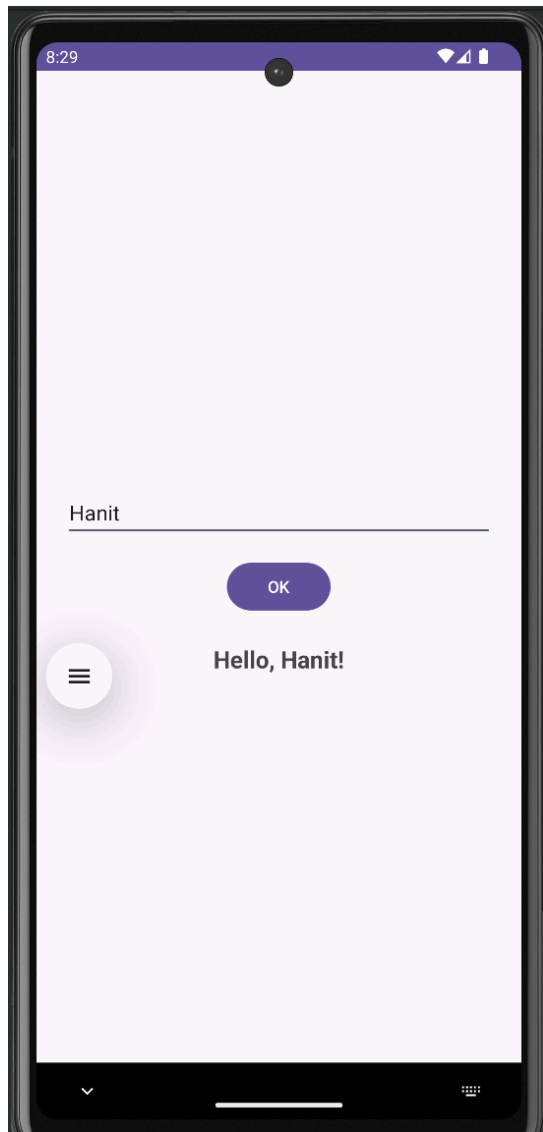
    EditText etName;
    Button btnOK;
    TextView tvHello;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etName = findViewById(R.id.etName);
        btnOK = findViewById(R.id.btnOK);
        tvHello = findViewById(R.id.tvHello);

        btnOK.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String name = etName.getText().toString().trim();
                if (!name.isEmpty()) {
                    tvHello.setText("Hello, " + name + "!");
                } else {
                    tvHello.setText("Please enter a name.");
                }
            }
        });
    }
}
```

Output -



Q2 -> Create a screen that has input boxes for User Name, Password, Address, Gender (radio buttons for male and female), Age (numeric), Date of Birth (Date Picket), State (Spinner) and a Submit button. On clicking the submit button, print all the data below the Submit Button. Use 1. Linear Layout, 2. Relative Lavout and 3. Grid Layout or Table Layout.

Ans2 ->

Java -

```
package com.example.formapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
```

```

import android.view.View;
import android.widget.*;

public class MainActivity extends AppCompatActivity {

    EditText etName, etPassword, etAddress, etAge;
    RadioGroup rgGender;
    DatePicker dpDOB;
    Spinner spState;
    Button btnSubmit;
    TextView tvResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main); // swap XML here

        etName      = findViewById(R.id.etName);
        etPassword   = findViewById(R.id.etPassword);
        etAddress    = findViewById(R.id.etAddress);
        etAge        = findViewById(R.id.etAge);
        rgGender     = findViewById(R.id.rgGender);
        dpDOB        = findViewById(R.id.dpDOB);
        spState      = findViewById(R.id.spState);
        btnSubmit    = findViewById(R.id.btnSubmit);
        tvResult     = findViewById(R.id.tvResult);

        // Populate spinner
        String[] states = {"Delhi", "Haryana", "UP", "Maharashtra",
"Karnataka"};
        ArrayAdapter<String> adapter = new ArrayAdapter<>(this,
            android.R.layout.simple_spinner_item, states);

        adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item);

        spState.setAdapter(adapter);

        btnSubmit.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String name      = etName.getText().toString();
                String password  = etPassword.getText().toString();
                String address   = etAddress.getText().toString();
                String age       = etAge.getText().toString();
                String state     = spState.getSelectedItem().toString();

                int selectedGenderId = rgGender.getCheckedRadioButtonId();
                RadioButton rbGender = findViewById(selectedGenderId);

```

```

        String gender = (rbGender != null) ?
rbGender.getText().toString() : "Not selected";

        String dob = dpDOB.getDayOfMonth() + "/" +
(dpDOB.getMonth() + 1) + "/" +
dpDOB.getYear();

        tvResult.setText(
            "Name: " + name + "\n" +
            "Password: " + password + "\n" +
            "Address: " + address + "\n" +
            "Gender: " + gender + "\n" +
            "Age: " + age + "\n" +
            "DOB: " + dob + "\n" +
            "State: " + state
        );
    }
});
}
}

```

Xml (Linear Layout) -

```

<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp">

        <TextView android:text="User Name" android:layout_width="wrap_content"
android:layout_height="wrap_content"/>
        <EditText android:id="@+id/etName" android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Enter name"/>

        <TextView android:text="Password" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:layout_marginTop="8dp"/>
        <EditText android:id="@+id/etPassword"
android:layout_width="match_parent" android:layout_height="wrap_content"
android:inputType="textPassword" android:hint="Enter password"/>

        <TextView android:text="Address" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:layout_marginTop="8dp"/>

```

```

        <EditText android:id="@+id/etAddress"
android:layout_width="match_parent" android:layout_height="wrap_content"
android:hint="Enter address"/>

        <TextView android:text="Gender" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:layout_marginTop="8dp"/>
        <RadioGroup android:id="@+id/rgGender" android:orientation="horizontal"
android:layout_width="wrap_content" android:layout_height="wrap_content">
            <RadioButton android:id="@+id/rbMale" android:text="Male"
android:layout_width="wrap_content" android:layout_height="wrap_content"/>
            <RadioButton android:id="@+id/rbFemale" android:text="Female"
android:layout_width="wrap_content" android:layout_height="wrap_content"/>
        </RadioGroup>

        <TextView android:text="Age" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:layout_marginTop="8dp"/>
        <EditText android:id="@+id/etAge" android:layout_width="match_parent"
android:layout_height="wrap_content" android:inputType="number"
android:hint="Enter age"/>

        <TextView android:text="Date of Birth"
android:layout_width="wrap_content" android:layout_height="wrap_content"
android:layout_marginTop="8dp"/>
        <DatePicker android:id="@+id/dpDOB" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:calendarViewShown="false"/>

        <TextView android:text="State" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:layout_marginTop="8dp"/>
        <Spinner android:id="@+id/spState" android:layout_width="match_parent"
android:layout_height="wrap_content"/>

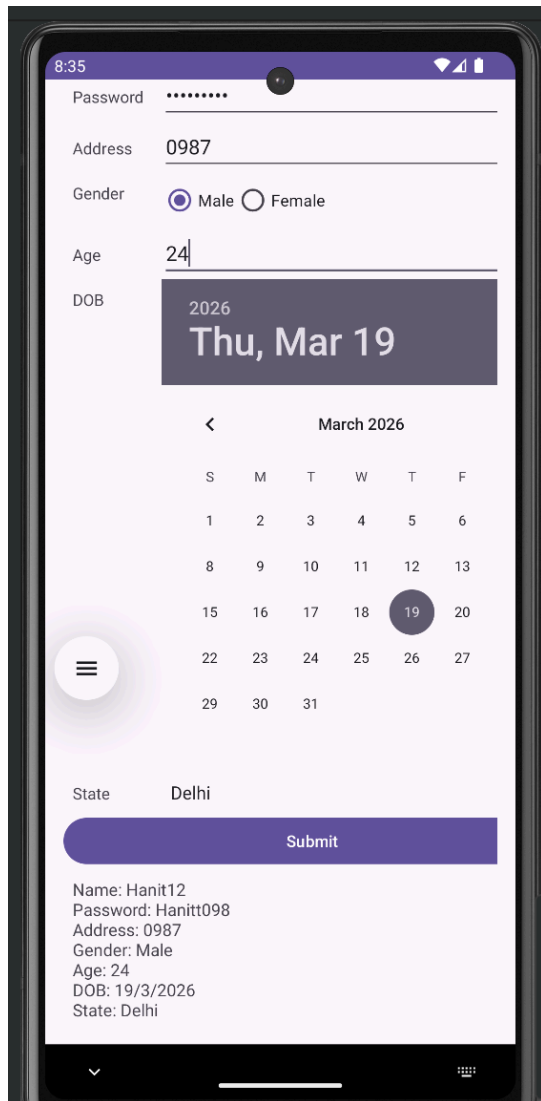
        <Button android:id="@+id/btnSubmit" android:layout_width="match_parent"
android:layout_height="wrap_content" android:layout_marginTop="16dp"
android:text="Submit"/>

        <TextView android:id="@+id/tvResult"
android:layout_width="wrap_content" android:layout_height="wrap_content"
android:layout_marginTop="16dp" android:textSize="15sp"/>

    </LinearLayout>
</ScrollView>

```

Output -



Xml (Relative Layout) -

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <RelativeLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:padding="16dp">

        <TextView android:id="@+id/lblName" android:text="User Name"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"/>
        <EditText android:id="@+id/etName" android:hint="Enter name"
```

```

        android:layout_width="match_parent"
android:layout_height="wrap_content"
        android:layout_below="@id/lblName"/>

        <TextView android:id="@+id/lblPassword" android:text="Password"
        android:layout_width="wrap_content"
android:layout_height="wrap_content"
        android:layout_below="@id/etName" android:layout_marginTop="8dp"/>
        <EditText android:id="@+id/etPassword" android:hint="Enter password"
        android:inputType="textPassword"
        android:layout_width="match_parent"
android:layout_height="wrap_content"
        android:layout_below="@id/lblPassword"/>

        <TextView android:id="@+id/lblAddress" android:text="Address"
        android:layout_width="wrap_content"
android:layout_height="wrap_content"
        android:layout_below="@id/etPassword"
android:layout_marginTop="8dp"/>
        <EditText android:id="@+id/etAddress" android:hint="Enter address"
        android:layout_width="match_parent"
android:layout_height="wrap_content"
        android:layout_below="@id/lblAddress"/>

        <TextView android:id="@+id/lblGender" android:text="Gender"
        android:layout_width="wrap_content"
android:layout_height="wrap_content"
        android:layout_below="@id/etAddress"
android:layout_marginTop="8dp"/>
        <RadioGroup android:id="@+id/rgGender" android:orientation="horizontal"
        android:layout_width="wrap_content"
android:layout_height="wrap_content"
        android:layout_below="@id/lblGender">
            <RadioButton android:id="@+id/rbMale" android:text="Male"
                android:layout_width="wrap_content"
android:layout_height="wrap_content"/>
            <RadioButton android:id="@+id/rbFemale" android:text="Female"
                android:layout_width="wrap_content"
android:layout_height="wrap_content"/>
        </RadioGroup>

        <TextView android:id="@+id/lblAge" android:text="Age"
        android:layout_width="wrap_content"
android:layout_height="wrap_content"
        android:layout_below="@id/rgGender"
android:layout_marginTop="8dp"/>
        <EditText android:id="@+id/etAge" android:hint="Enter age"
        android:inputType="number"

```



```

        android:layout_width="match_parent"
android:layout_height="wrap_content"
        android:layout_below="@id/lblAge"/>

        <TextView android:id="@+id/lblDOB" android:text="Date of Birth"
        android:layout_width="wrap_content"
android:layout_height="wrap_content"
        android:layout_below="@id/etAge" android:layout_marginTop="8dp"/>
        <DatePicker android:id="@+id/dpDOB"
        android:layout_width="wrap_content"
android:layout_height="wrap_content"
        android:calendarViewShown="false"
        android:layout_below="@id/lblDOB"/>

        <TextView android:id="@+id/lblState" android:text="State"
        android:layout_width="wrap_content"
android:layout_height="wrap_content"
        android:layout_below="@id/dpDOB" android:layout_marginTop="8dp"/>
        <Spinner android:id="@+id/spState"
        android:layout_width="match_parent"
android:layout_height="wrap_content"
        android:layout_below="@id/lblState"/>

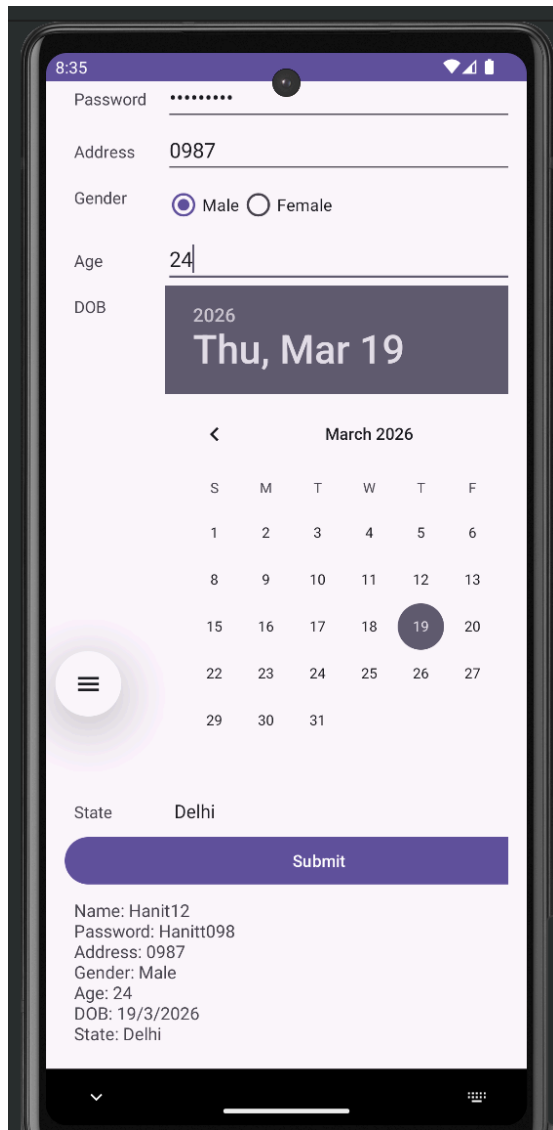
        <Button android:id="@+id/btnSubmit" android:text="Submit"
        android:layout_width="match_parent"
android:layout_height="wrap_content"
        android:layout_below="@id/spState"
android:layout_marginTop="16dp"/>

        <TextView android:id="@+id/tvResult"
        android:layout_width="wrap_content"
android:layout_height="wrap_content"
        android:layout_below="@id/btnSubmit"
android:layout_marginTop="16dp"
        android:textSize="15sp"/>

    </RelativeLayout>
</ScrollView>

```

Output -



Xml (Table Layout) -

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TableLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:padding="16dp"
        android:stretchColumns="1">

        <TableRow>
            <TextView android:text="User Name" android:padding="8dp"/>
```

```

        <EditText android:id="@+id/etName" android:hint="Enter name"/>
    </TableRow>

    <TableRow>
        <TextView android:text="Password" android:padding="8dp"/>
        <EditText android:id="@+id/etPassword" android:hint="Enter
password" android:inputType="textPassword"/>
    </TableRow>

    <TableRow>
        <TextView android:text="Address" android:padding="8dp"/>
        <EditText android:id="@+id/etAddress" android:hint="Enter
address"/>
    </TableRow>

    <TableRow>
        <TextView android:text="Gender" android:padding="8dp"/>
        <RadioGroup android:id="@+id/rgGender"
android:orientation="horizontal">
            <RadioButton android:id="@+id/rbMale" android:text="Male"/>
            <RadioButton android:id="@+id/rbFemale" android:text="Female"/>
        </RadioGroup>
    </TableRow>

    <TableRow>
        <TextView android:text="Age" android:padding="8dp"/>
        <EditText android:id="@+id/etAge" android:hint="Enter age"
android:inputType="number"/>
    </TableRow>

    <TableRow>
        <TextView android:text="DOB" android:padding="8dp"/>
        <DatePicker android:id="@+id/dpDOB"
android:calendarViewShown="false"/>
    </TableRow>

    <TableRow>
        <TextView android:text="State" android:padding="8dp"/>
        <Spinner android:id="@+id/spState"/>
    </TableRow>

    <TableRow>
        <Button android:id="@+id/btnSubmit" android:text="Submit"
            android:layout_span="2" android:layout_width="match_parent"/>
    </TableRow>

    <TableRow>
        <TextView android:id="@+id/tvResult" android:layout_span="2"
            android:padding="8dp" android:textSize="15sp"/>
    </TableRow>

```

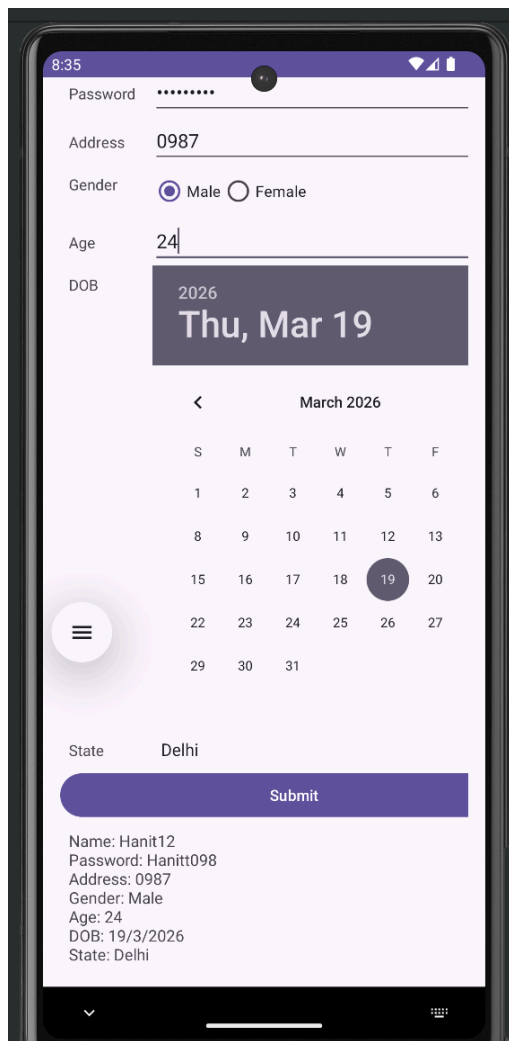
```

        </TableRow>

    </TableLayout>
</ScrollView>

```

Output -



Q3 -> Develop an application that inserts some notifications into Notification area and whenever a notification is inserted, it should show a toast with details of the notification.

Ans3 ->

Xml -

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

```

```

        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:gravity="center"
        android:padding="24dp">

        <EditText
            android:id="@+id/etTitle"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Notification Title"/>

        <EditText
            android:id="@+id/etMessage"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Notification Message"
            android:layout_marginTop="12dp"/>

        <Button
            android:id="@+id/btnSend"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginTop="20dp"
            android:text="Send Notification"/>

</LinearLayout>

```

Java -

```

package com.example.q3notification;

import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.NotificationCompat;
import androidx.core.app.NotificationManagerCompat;
import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.os.Build;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    EditText etTitle, etMessage;
    Button btnSend;

    private static final String CHANNEL_ID = "my_channel";

```

```

private static int notificationId = 0;

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    etTitle = findViewById(R.id.etTitle);
    etMessage = findViewById(R.id.etMessage);
    btnSend = findViewById(R.id.btnSend);

    createNotificationChannel();

    btnSend.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            String title = etTitle.getText().toString().trim();
            String message = etMessage.getText().toString().trim();

            if (title.isEmpty() || message.isEmpty()) {
                Toast.makeText(MainActivity.this,
                    "Please enter title and message",
                    Toast.LENGTH_SHORT).show();
                return;
            }

            // Build and show notification
            NotificationCompat.Builder builder = new
NotificationCompat.Builder(
                MainActivity.this, CHANNEL_ID)
                .setSmallIcon(android.R.drawable.ic_dialog_info)
                .setContentTitle(title)
                .setContentText(message)
                .setPriority(NotificationCompat.PRIORITY_HIGH)
                .setAutoCancel(true);

            NotificationManagerCompat notificationManager =
                NotificationManagerCompat.from(MainActivity.this);
            notificationManager.notify(notificationId++, builder.build());

            // Show Toast with notification details
            Toast.makeText(MainActivity.this,
                "Notification Sent!\nTitle: " + title + "\nMessage: " +
message,
                Toast.LENGTH_LONG).show();
        }
    });
}

```

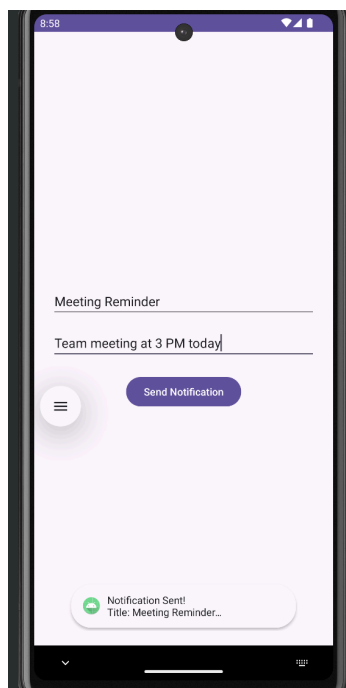
```

private void createNotificationChannel() {
    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
        CharSequence name = "My Channel";
        String description = "Channel for app notifications";
        int importance = NotificationManager.IMPORTANCE_HIGH;
        NotificationChannel channel = new NotificationChannel(CHANNEL_ID,
name, importance);
        channel.setDescription(description);

        NotificationManager notificationManager =
            getSystemService(NotificationManager.class);
        notificationManager.createNotificationChannel(channel);
    }
}
}

```

Output -



Q4 -> Develop an application that uses a menu with 3 options for dialing a number, opening a website and to send an SMS. On selecting an option, the appropriate action should be invoked using intents.

Ans4 ->

Xml -

```

<?xml version="1.0" encoding="utf-8"?>

```

```

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <androidx.appcompat.widget.Toolbar
        android:id="@+id/toolbar"
        android:layout_width="match_parent"
        android:layout_height="?attr/actionBarSize"
        android:background="?attr/colorPrimary"
        android:title="Q4 Menu App"/>

    <TextView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:gravity="center"
        android:text="Press the 3-dot menu (top right)"
        android:textSize="18sp"/>

</LinearLayout>

```

Java -

```

package com.example.q4menu;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.Toast;
import androidx.appcompat.widget.Toolbar;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Toolbar toolbar = findViewById(R.id.toolbar);
        setSupportActionBar(toolbar);
    }

    // Inflate the menu
    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.menu_main, menu);
        return true;
    }
}

```



```

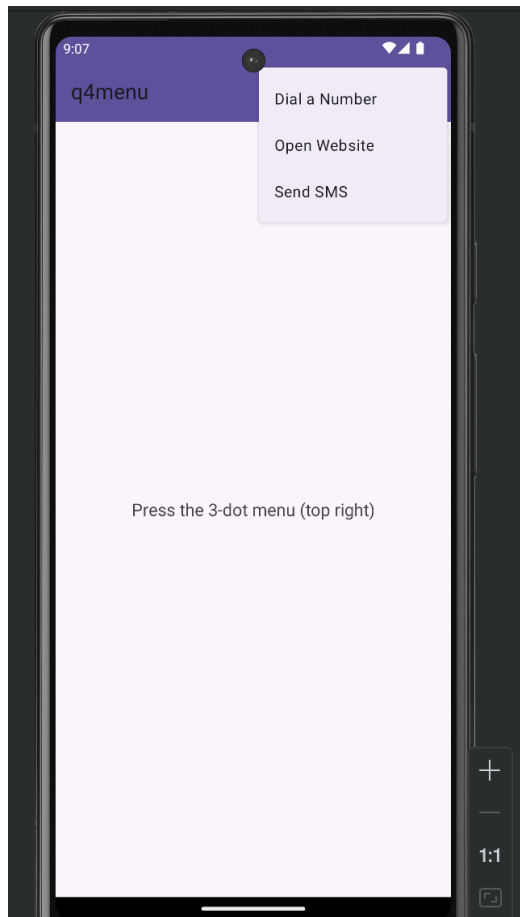
// Handle menu item clicks
@Override
public boolean onOptionsItemSelected(MenuItem item) {
    int id = item.getItemId();

    if (id == R.id.menu_dial) {
        // Dial a number
        Intent dialIntent = new Intent(Intent.ACTION_DIAL);
        dialIntent.setData(Uri.parse("tel:9876543210"));
        startActivity(dialIntent);
        return true;
    } else if (id == R.id.menu_website) {
        // Open a website
        Intent webIntent = new Intent(Intent.ACTION_VIEW);
        webIntent.setData(Uri.parse("https://www.google.com"));
        startActivity(webIntent);
        return true;
    } else if (id == R.id.menu_sms) {
        // Send an SMS
        Intent smsIntent = new Intent(Intent.ACTION_SENDTO);
        smsIntent.setData(Uri.parse("smsto:9876543210"));
        smsIntent.putExtra("sms_body", "Hello! This is a test SMS.");
        startActivity(smsIntent);
        return true;
    }

    return super.onOptionsItemSelected(item);
}
}

```

Output -



Q5 -> Create an application that uses a text file to store user names and passwords (tab separated fields and one record per line) When the user submits a login name and password through a screen, the details should be verified with the text file data and if they match, show a dialog saying that login is successful. Otherwise, show the dialog with Login Failed message.

Ans5 ->

Xml -

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">

    <TextView
        android:layout_width="wrap_content"
```

```

        android:layout_height="wrap_content"
        android:text="Login"
        android:textSize="28sp"
        android:textStyle="bold"
        android:layout_marginBottom="24dp"/>

<EditText
    android:id="@+id/etUsername"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Username"
    android:inputType="textPersonName"/>

<EditText
    android:id="@+id/etPassword"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Password"
    android:inputType="textPassword"
    android:layout_marginTop="12dp"/>

<Button
    android:id="@+id/btnLogin"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="20dp"
    android:text="Login"/>

<Button
    android:id="@+id/btnRegister"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="8dp"
    android:text="Register (Save User)"/>

</LinearLayout>

```

Java -

```

package com.example.q5login;

import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import java.io.*;

```

```

public class MainActivity extends AppCompatActivity {

    EditText etUsername, etPassword;
    Button btnLogin, btnRegister;
    private static final String FILE_NAME = "users.txt";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etUsername = findViewById(R.id.etUsername);
        etPassword = findViewById(R.id.etPassword);
        btnLogin = findViewById(R.id.btnLogin);
        btnRegister = findViewById(R.id.btnRegister);

        // Save new user to text file
        btnRegister.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String username = etUsername.getText().toString().trim();
                String password = etPassword.getText().toString().trim();

                if (username.isEmpty() || password.isEmpty()) {
                    Toast.makeText(MainActivity.this,
                        "Please enter username and password",
                        Toast.LENGTH_SHORT).show();
                    return;
                }

                try {
                    FileOutputStream fos = openFileOutput(FILE_NAME,
                        MODE_APPEND);

                    String record = username + "\t" + password + "\n";
                    fos.write(record.getBytes());
                    fos.close();
                    Toast.makeText(MainActivity.this,
                        "User registered successfully!",
                        Toast.LENGTH_SHORT).show();
                } catch (IOException e) {
                    e.printStackTrace();
                }
            }
        });

        // Verify login from text file
        btnLogin.setOnClickListener(new View.OnClickListener() {
            @Override

```

```

        public void onClick(View v) {
            String username = etUsername.getText().toString().trim();
            String password = etPassword.getText().toString().trim();

            if (username.isEmpty() || password.isEmpty()) {
                Toast.makeText(MainActivity.this,
                    "Please enter username and password",
                    Toast.LENGTH_SHORT).show();
                return;
            }

            boolean loginSuccess = false;

            try {
                FileInputStream fis = openFileInput(FILE_NAME);
                BufferedReader reader = new BufferedReader(new
                    InputStreamReader(fis));
                String line;

                while ((line = reader.readLine()) != null) {
                    String[] parts = line.split("\t");
                    if (parts.length == 2) {
                        String savedUser = parts[0].trim();
                        String savedPass = parts[1].trim();
                        if (savedUser.equals(username) &&
                            savedPass.equals(password)) {
                            loginSuccess = true;
                            break;
                        }
                    }
                }
                reader.close();
                fis.close();

            } catch (FileNotFoundException e) {
                showDialog("Login Failed", "No users registered yet!");
                return;
            } catch (IOException e) {
                e.printStackTrace();
            }

            if (loginSuccess) {
                showDialog("Login Successful", "Welcome, " + username +
                    "!");
            } else {
                showDialog("Login Failed", "Invalid username or
                    password.");
            }
        }
    }

```

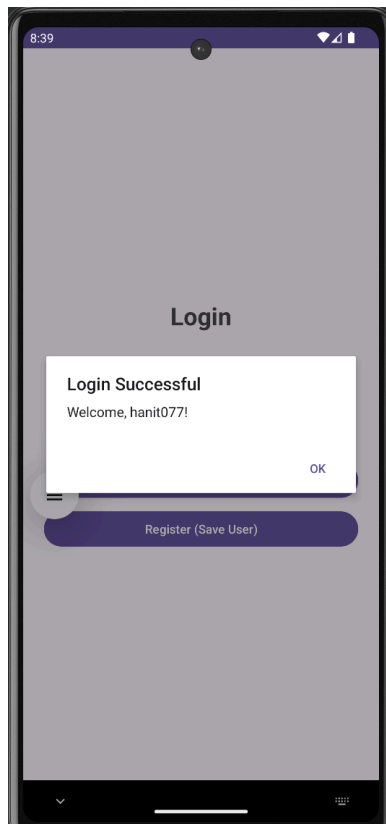
```

    });
}

private void showDialog(String title, String message) {
    new AlertDialog.Builder(this)
        .setTitle(title)
        .setMessage(message)
        .setPositiveButton("OK", null)
        .show();
}
}

```

Output -



Q6 -> Create an activity to navigate to different activities through Intents-Implicit and Explicit.

Ans6 ->

Xml -

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"

```

```

        android:layout_height="match_parent"
        android:orientation="vertical"
        android:gravity="center"
        android:padding="24dp">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Intent Navigation Demo"
            android:textSize="22sp"
            android:textStyle="bold"
            android:layout_marginBottom="32dp"/>

        <Button
            android:id="@+id/btnExplicit"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Explicit Intent (Go to Second Activity)"/>

        <Button
            android:id="@+id/btnImplicitWeb"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginTop="12dp"
            android:text="Implicit Intent (Open Website)"/>

        <Button
            android:id="@+id/btnImplicitCall"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginTop="12dp"
            android:text="Implicit Intent (Dial Number)"/>

        <Button
            android:id="@+id/btnImplicitMap"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginTop="12dp"
            android:text="Implicit Intent (Open Map)"/>

    </LinearLayout>

```

Xml (activity_second.xml) -

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"

```

```

        android:orientation="vertical"
        android:gravity="center"
        android:padding="24dp">

        <TextView
            android:id="@+id/tvMessage"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Welcome to Second Activity!"
            android:textSize="22sp"
            android:textStyle="bold"/>

        <Button
            android:id="@+id/btnBack"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginTop="24dp"
            android:text="Go Back"/>

    </LinearLayout>

```

Java (MainActivity.java) -

```

package com.example.q6intents;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {

    Button btnExplicit, btnImplicitWeb, btnImplicitCall, btnImplicitMap;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnExplicit = findViewById(R.id.btnExplicit);
        btnImplicitWeb = findViewById(R.id.btnImplicitWeb);
        btnImplicitCall = findViewById(R.id.btnImplicitCall);
        btnImplicitMap = findViewById(R.id.btnImplicitMap);

        // EXPLICIT INTENT - go to SecondActivity
        btnExplicit.setOnClickListener(new View.OnClickListener() {
            @Override

```



```

        public void onClick(View v) {
            Intent intent = new Intent(MainActivity.this,
SecondActivity.class);
            intent.putExtra("message", "Hello from MainActivity!");
            startActivity(intent);
        }
    });

    // IMPLICIT INTENT - open website
    btnImplicitWeb.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            Intent intent = new Intent(Intent.ACTION_VIEW);
            intent.setData(Uri.parse("https://www.google.com"));
            startActivity(intent);
        }
    });

    // IMPLICIT INTENT - dial number
    btnImplicitCall.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            Intent intent = new Intent(Intent.ACTION_DIAL);
            intent.setData(Uri.parse("tel:9876543210"));
            startActivity(intent);
        }
    });

    // IMPLICIT INTENT - open map
    btnImplicitMap.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            Intent intent = new Intent(Intent.ACTION_VIEW);
            intent.setData(Uri.parse("geo:28.6139,77.2090?q=New+Delhi"));
            startActivity(intent);
        }
    });
}
}

```

Java (SecondActivity.java) -

```

package com.example.q6intents;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;

```

```

public class SecondActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);

        TextView tvMessage = findViewById(R.id.tvMessage);
        Button btnBack = findViewById(R.id.btnBack);

        // Receive data from MainActivity
        String message = getIntent().getStringExtra("message");
        if (message != null) {
            tvMessage.setText(message);
        }

        btnBack.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                finish(); // goes back to MainActivity
            }
        });
    }
}

```

Output -



Q7 -> Create an activity to create option Menu.

Ans7 ->

Xml (activity_main.xml) -

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Select an option from the menu"
        android:textSize="20sp"
        android:gravity="center"
        android:textStyle="bold"/>

</LinearLayout>
```

Xml (menu_options.xml) -

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">

    <item
        android:id="@+id/menu_home"
        android:title="Home"
        android:icon="@android:drawable/ic_menu_today"
        android:showAsAction="ifRoom"/>

    <item
        android:id="@+id/menu_search"
        android:title="Search"
        android:icon="@android:drawable/ic_menu_search"
        android:showAsAction="ifRoom"/>

    <item
        android:id="@+id/menu_settings"
        android:title="Settings"
        android:icon="@android:drawable/ic_menu_preferences"
        android:showAsAction="never"/>

    <item
        android:id="@+id/menu_share"
        android:title="Share"
        android:icon="@android:drawable/ic_menu_share"
        android:showAsAction="never"/>

    <item
        android:id="@+id/menu_about"
        android:title="About"
        android:icon="@android:drawable/ic_menu_info_details"
        android:showAsAction="never"/>

    <item
        android:id="@+id/menu_exit"
        android:title="Exit"
        android:icon="@android:drawable/ic_menu_close_clear_cancel"
        android:showAsAction="never"/>

</menu>
```

Java -

```
package com.example.q7optionmenu;

import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
```

```
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.TextView;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    TextView tvResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        tvResult = findViewById(R.id.tvResult);
    }

    // Inflate the menu
    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.menu_options, menu);
        return true;
    }

    // Handle menu item clicks
    @Override
    public boolean onOptionsItemSelected(MenuItem item) {
        int id = item.getItemId();

        if (id == R.id.menu_home) {
            tvResult.setText("You selected: Home");
            Toast.makeText(this, "Home selected", Toast.LENGTH_SHORT).show();
            return true;
        } else if (id == R.id.menu_search) {
            tvResult.setText("You selected: Search");
            Toast.makeText(this, "Search selected", Toast.LENGTH_SHORT).show();
            return true;
        } else if (id == R.id.menu_settings) {
            tvResult.setText("You selected: Settings");
            Toast.makeText(this, "Settings selected",
Toast.LENGTH_SHORT).show();
            return true;
        } else if (id == R.id.menu_share) {
            tvResult.setText("You selected: Share");
            Toast.makeText(this, "Share selected", Toast.LENGTH_SHORT).show();
        }
    }
}
```

```

        return true;

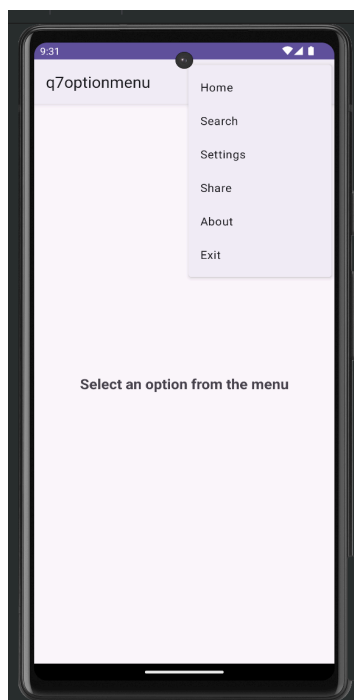
    } else if (id == R.id.menu_about) {
        // Show About dialog
        new AlertDialog.Builder(this)
            .setTitle("About")
            .setMessage("This is Question 7 - Option Menu
App\nDeveloped for Android Lab")
            .setPositiveButton("OK", null)
            .show();
        return true;

    } else if (id == R.id.menu_exit) {
        // Show exit confirmation dialog
        new AlertDialog.Builder(this)
            .setTitle("Exit")
            .setMessage("Are you sure you want to exit?")
            .setPositiveButton("Yes", (dialog, which) -> finish())
            .setNegativeButton("No", null)
            .show();
        return true;
    }

    return super.onOptionsItemSelected(item);
}
}

```

Output -



Q8 -> Create an activity using Button Controls- CheckBox, RadioButtons, Radio Groups
Ans8 ->

Xml -

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="24dp">

        <!-- CHECKBOXES -->
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Select Hobbies (CheckBox):"
            android:textSize="18sp"
            android:textStyle="bold"/>

        <CheckBox
            android:id="@+id/cbReading"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Reading"/>

        <CheckBox
            android:id="@+id/cbCoding"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Coding"/>

        <CheckBox
            android:id="@+id/cbGaming"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Gaming"/>

        <CheckBox
            android:id="@+id/cbTravelling"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Travelling"/>

        <!-- RADIO BUTTONS -->
        <TextView
```

```

        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Select Gender (RadioButton):"
        android:textSize="18sp"
        android:textStyle="bold"
        android:layout_marginTop="20dp"/>

<RadioGroup
    android:id="@+id/rgGender"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="horizontal">

    <RadioButton
        android:id="@+id/rbMale"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Male"/>

    <RadioButton
        android:id="@+id/rbFemale"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Female"/>

    <RadioButton
        android:id="@+id/rbOther"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Other"/>

</RadioGroup>

<!-- RADIO GROUP - Age Range -->
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Select Age Range (RadioGroup):"
    android:textSize="18sp"
    android:textStyle="bold"
    android:layout_marginTop="20dp"/>

<RadioGroup
    android:id="@+id/rgAge"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="vertical">

    <RadioButton

```



```

        android:id="@+id/rbAge1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Below 18"/>

<RadioButton
    android:id="@+id/rbAge2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="18 - 25"/>

<RadioButton
    android:id="@+id/rbAge3"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="26 - 35"/>

<RadioButton
    android:id="@+id/rbAge4"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Above 35"/>

</RadioGroup>

<!-- SUBMIT BUTTON -->
<Button
    android:id="@+id/btnSubmit"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="24dp"
    android:text="Submit"/>

<!-- RESULT -->
<TextView
    android:id="@+id/tvResult"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="16dp"
    android:textSize="16sp"
    android:textStyle="bold"/>

</LinearLayout>
</ScrollView>

```

Java -

```

package com.example.q8controls;

import androidx.appcompat.app.AppCompatActivity;

```

```

import android.os.Bundle;
import android.view.View;
import android.widget.*;

public class MainActivity extends AppCompatActivity {

    CheckBox cbReading, cbCoding, cbGaming, cbTravelling;
    RadioGroup rgGender, rgAge;
    Button btnSubmit;
    TextView tvResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        cbReading    = findViewById(R.id.cbReading);
        cbCoding      = findViewById(R.id.cbCoding);
        cbGaming      = findViewById(R.id.cbGaming);
        cbTravelling  = findViewById(R.id.cbTravelling);
        rgGender      = findViewById(R.id.rgGender);
        rgAge         = findViewById(R.id.rgAge);
        btnSubmit     = findViewById(R.id.btnSubmit);
        tvResult      = findViewById(R.id.tvResult);

        btnSubmit.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {

                // Get selected hobbies
                String hobbies = "";
                if (cbReading.isChecked()) hobbies += "Reading, ";
                if (cbCoding.isChecked()) hobbies += "Coding, ";
                if (cbGaming.isChecked()) hobbies += "Gaming, ";
                if (cbTravelling.isChecked()) hobbies += "Travelling, ";
                if (hobbies.isEmpty()) hobbies = "None";
                else hobbies = hobbies.substring(0, hobbies.length() - 2);

                // Get selected gender
                int genderId = rgGender.getCheckedRadioButtonId();
                String gender = "Not selected";
                if (genderId != -1) {
                    RadioButton rb = findViewById(genderId);
                    gender = rb.getText().toString();
                }

                // Get selected age range
                int ageId = rgAge.getCheckedRadioButtonId();
                String ageRange = "Not selected";
            }
        });
    }
}

```

```

        if (ageId != -1) {
            RadioButton rb = findViewById(ageId);
            ageRange = rb.getText().toString();
        }

        // Display result
        tvResult.setText(
            "Hobbies: " + hobbies + "\n" +
            "Gender: " + gender + "\n" +
            "Age Range: " + ageRange
        );
    }
}
}

```

Output -

